Cylinders and Quadric Surfaces

Outline:
1. Recognizing and graphing cylinders and quadric surfaces
2. 3D curves of intersection

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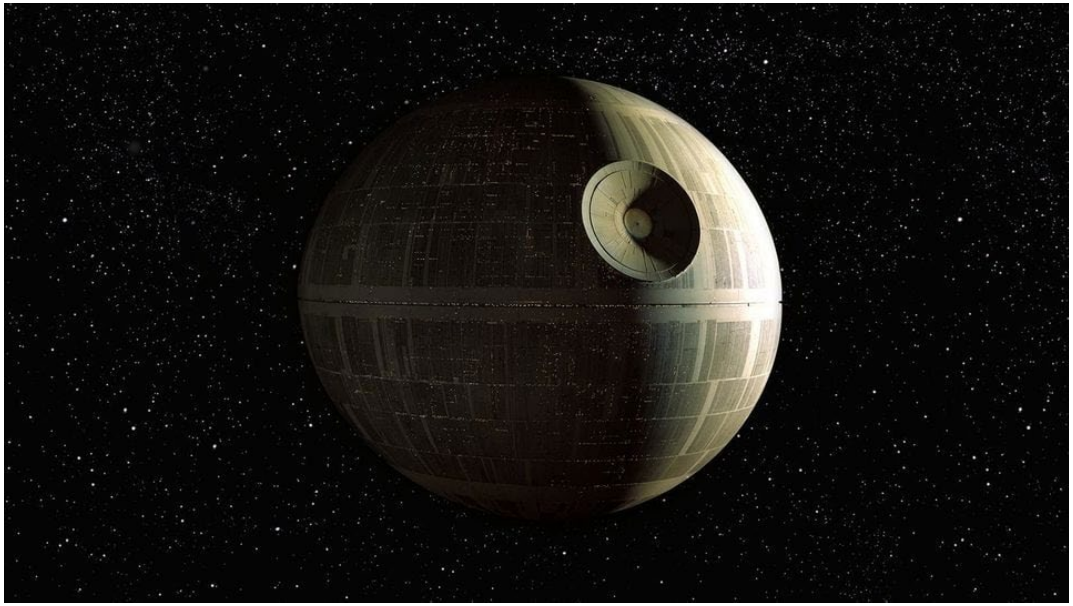
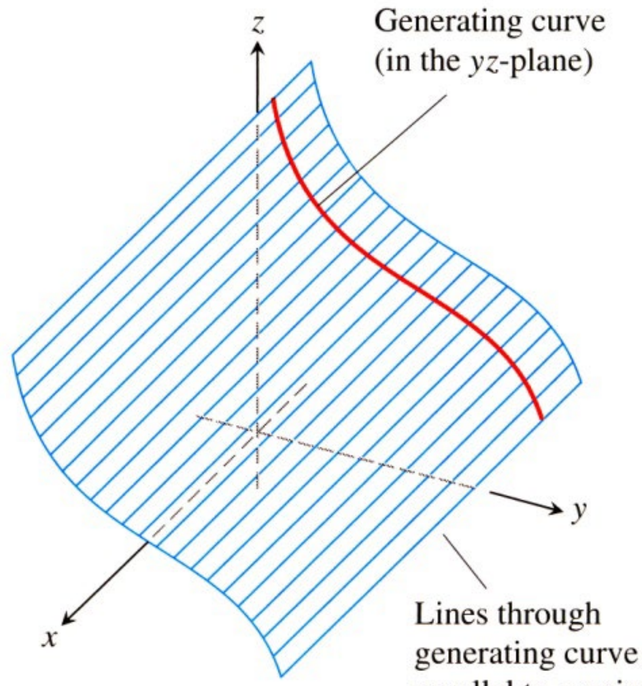


The Death Star was the Empire's ultimate weapon: a moon-sized space station with the ability to destroy an entire planet …

https://www.starwars.com/databank/death-star

Cylinders

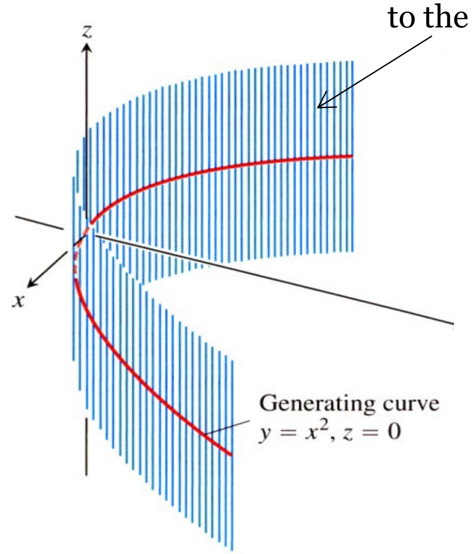


A cylinder is a surface generated by moving a straight line along a given planar curve while holding the line parallel to a given fixed line

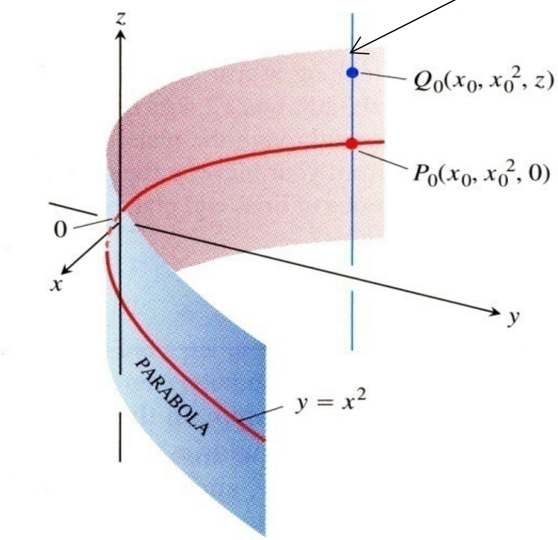
Cylinders

Equation: $y = x^2$ → Parabolic Cylinder

Lines parallel to the z – axis



Line through P_0 parallel to the z-axis



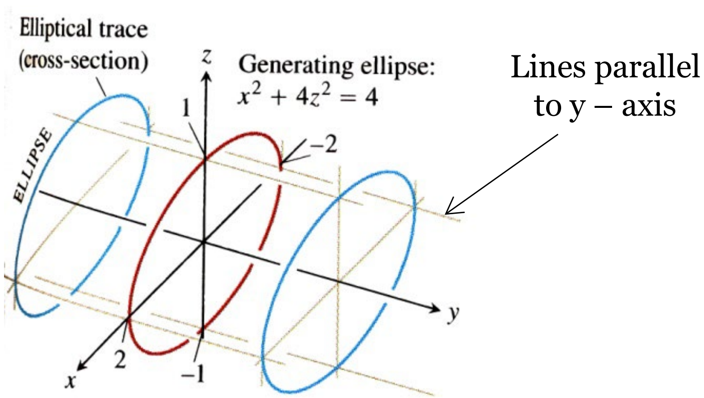
Every point in the cylinder has coordinates:

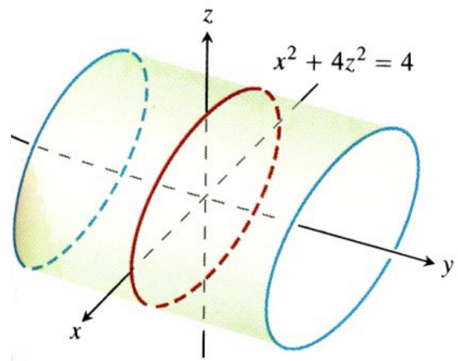
$$(x_0, x_0^2, z)$$

Cylinders

General Form: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ → Elliptical Cylinder

Example: $x^2 + 4z^2 = 4$





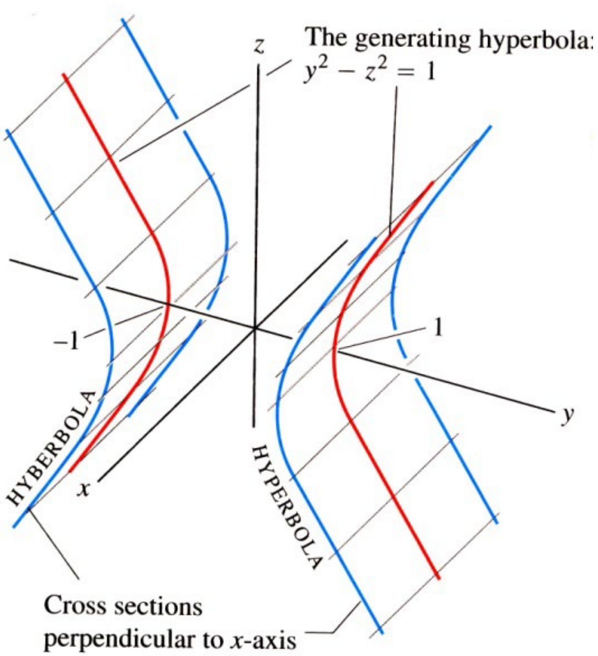
Every point in the cylinder has coordinates:

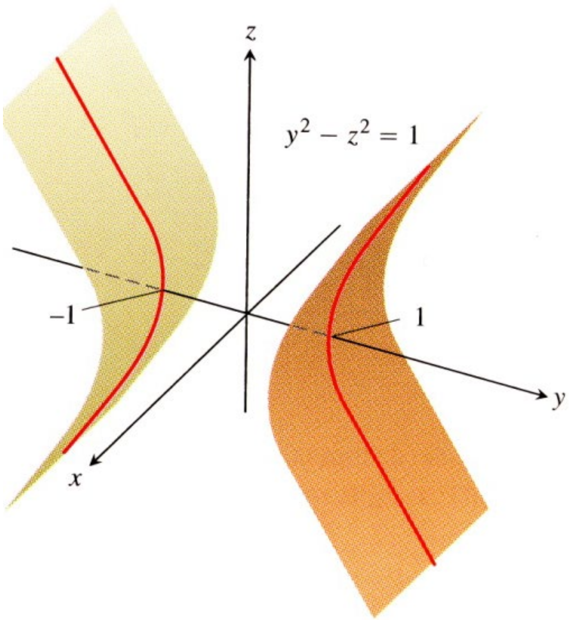
$$(x_0, y, z_0)$$

Cylinders

General Form: $\frac{y^2}{a^2} - \frac{z^2}{b^2} = 1$

Example: $y^2 - z^2 = 1$ → Hyperbolic Cylinder





Every point in the cylinder has coordinates:

(x, y_0, z_0)

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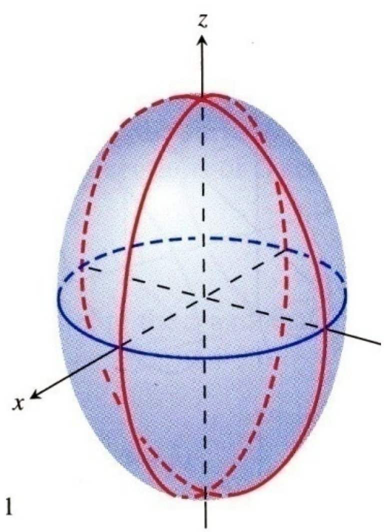
Quadric Surfaces

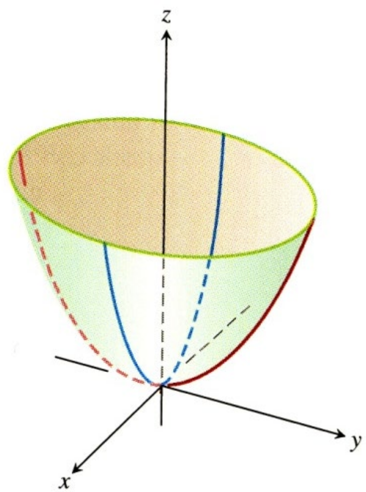
Graph of any equation that can be put into the general form:

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$$

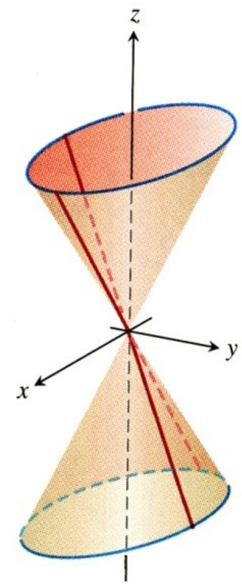
Where, A, B, …, are constants



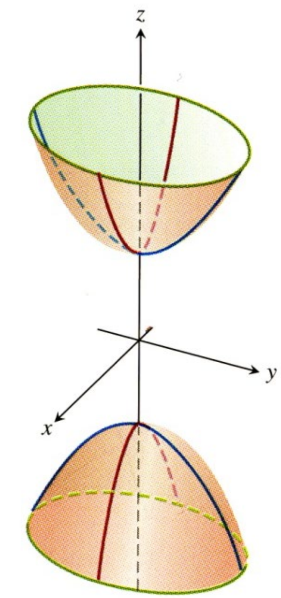
Ellipsoid



Paraboloid



Elliptical Cones



Hyperboloids

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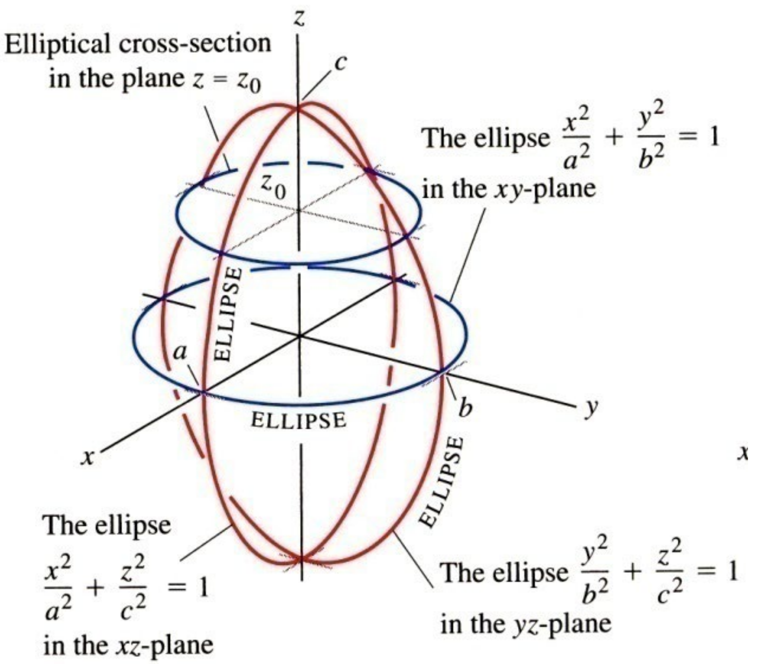
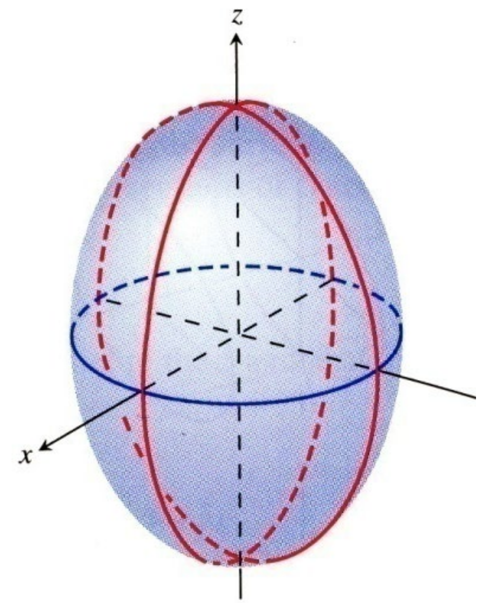
Quadric Surfaces

General Form: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$ → Ellipsoid

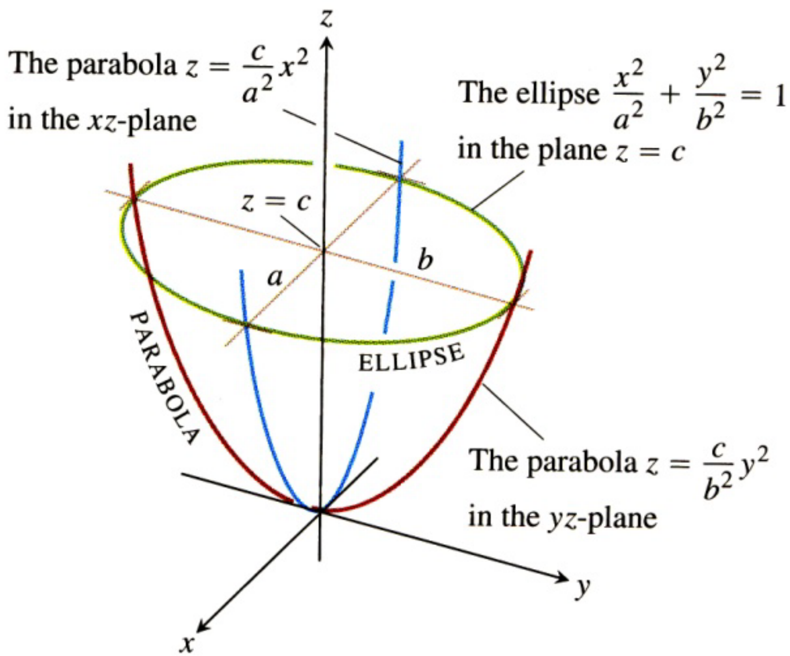


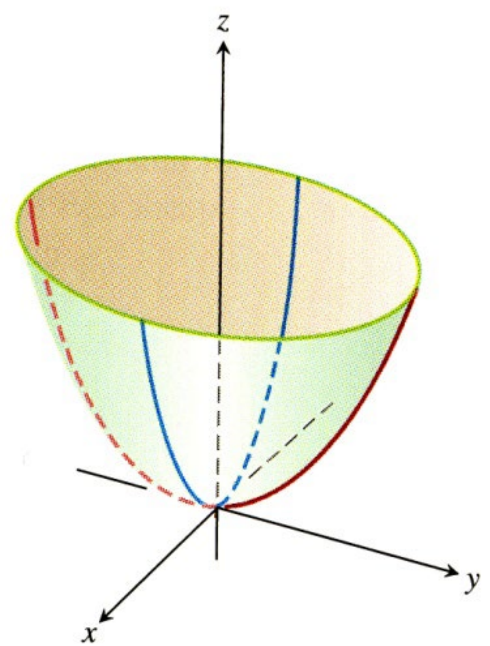


Quadric Surfaces

General Form: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = \dfrac{z}{c}$ → Elliptical Paraboloid

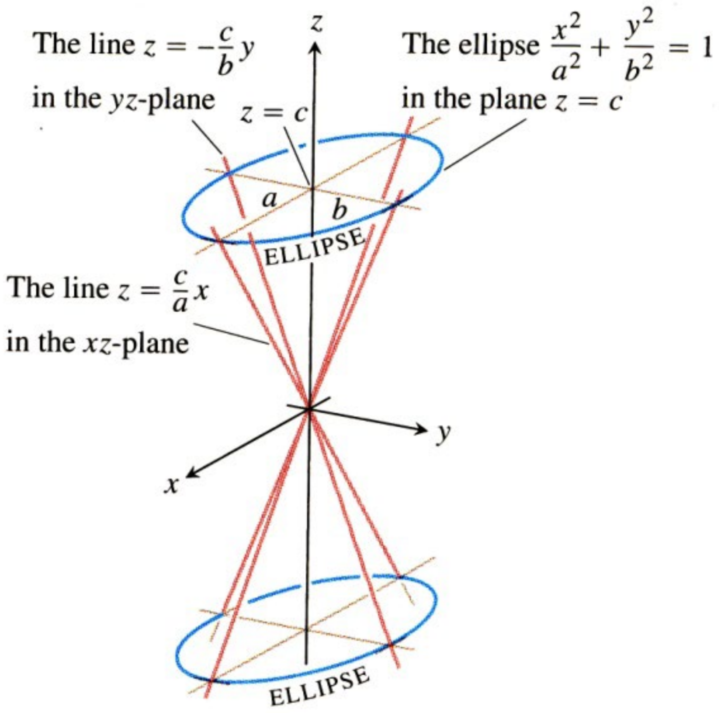
$c > 0$

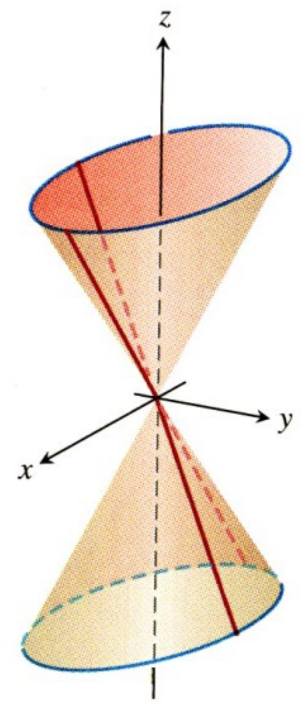




Quadric Surfaces

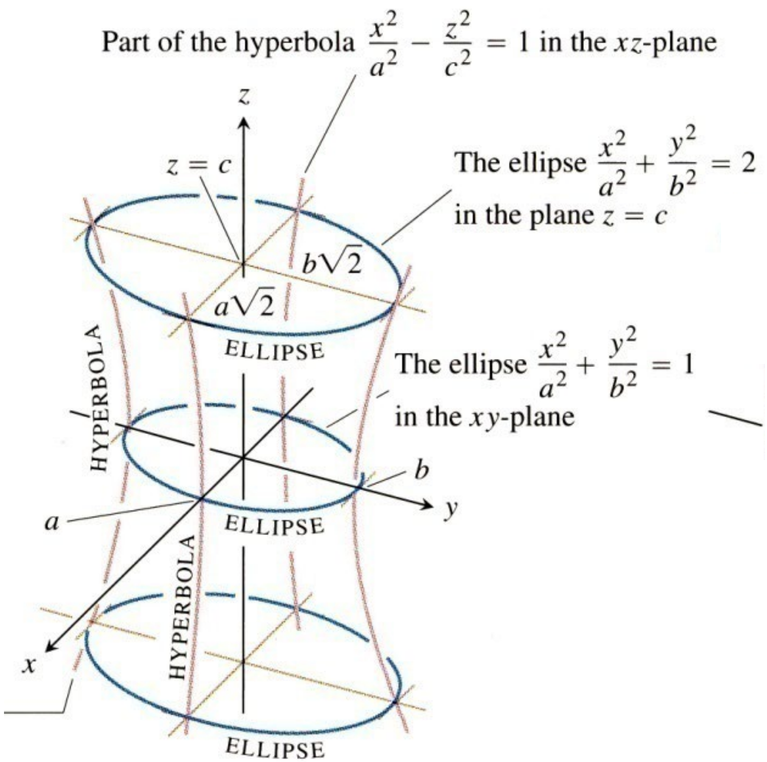
General Form: $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=\dfrac{z^2}{c^2}$ → Elliptical Cone

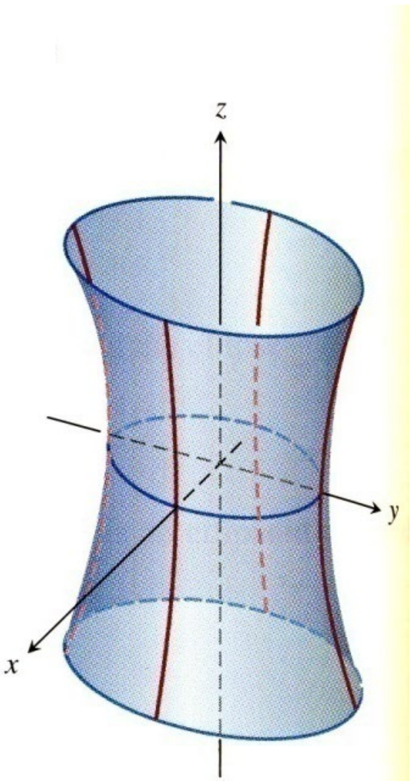




Quadric Surfaces

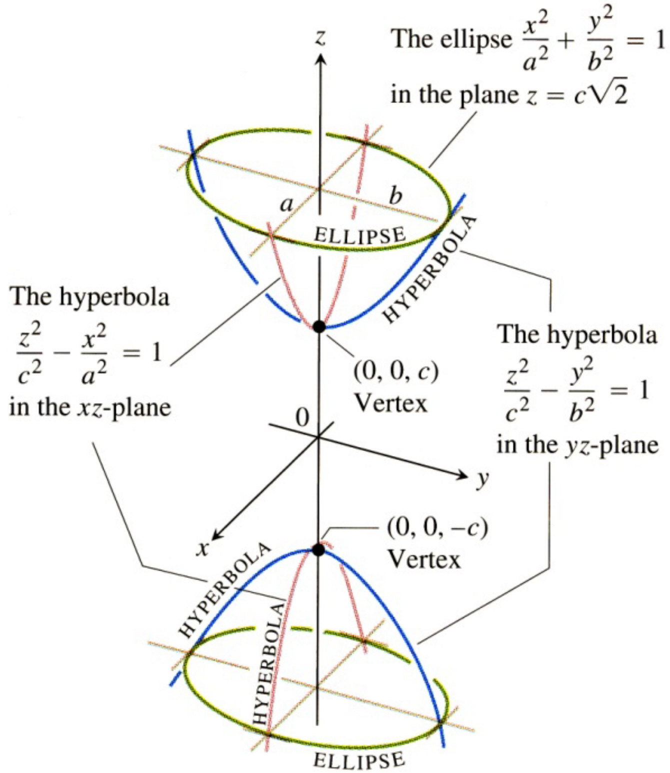
General Form: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$ → Hyperboloid of one sheet

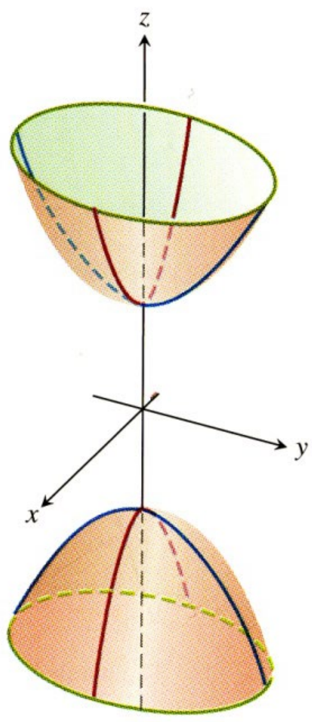




Quadric Surfaces

General Form: $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ → Hyperboloid of two sheets





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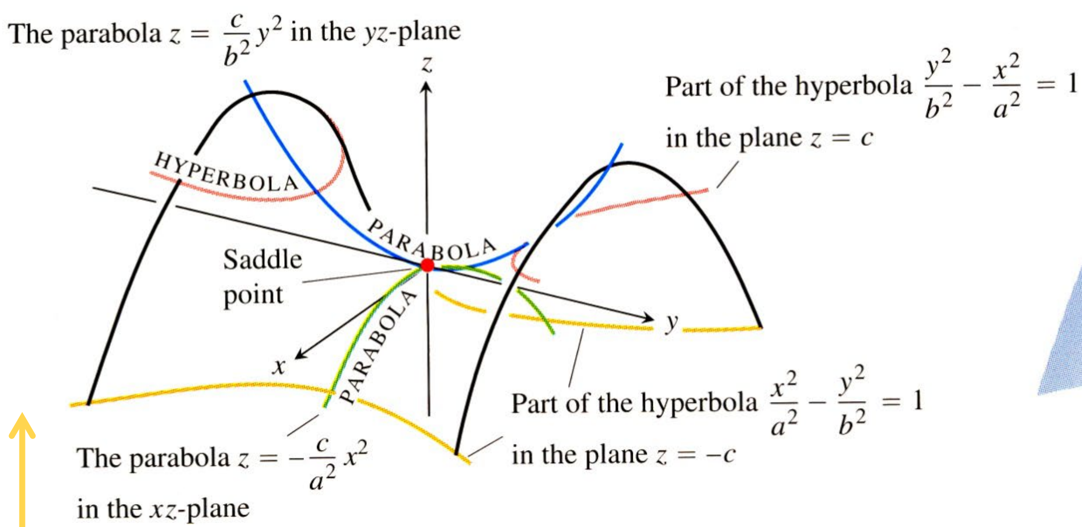
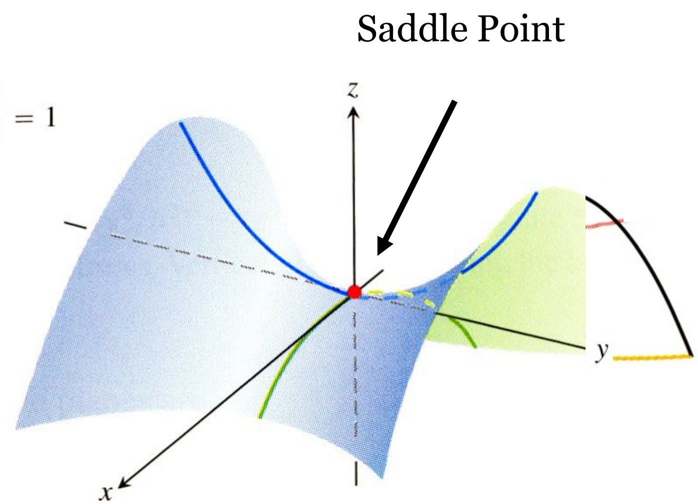
Quadric Surfaces

General Form: $\dfrac{y^2}{b^2} - \dfrac{x^2}{a^2} = \dfrac{z}{c}$ → Hyperbolic Paraboloid





$c > 0$

3D Curves of Intersection

Problem

A particle moves along the curve represented by the intersection of the surfaces

$$z = x^2 + y^2$$

and,

$$x + z = 1.$$

Draw the curve of intersection and find the equation for the projection of the curve in the xy plane

Solution:

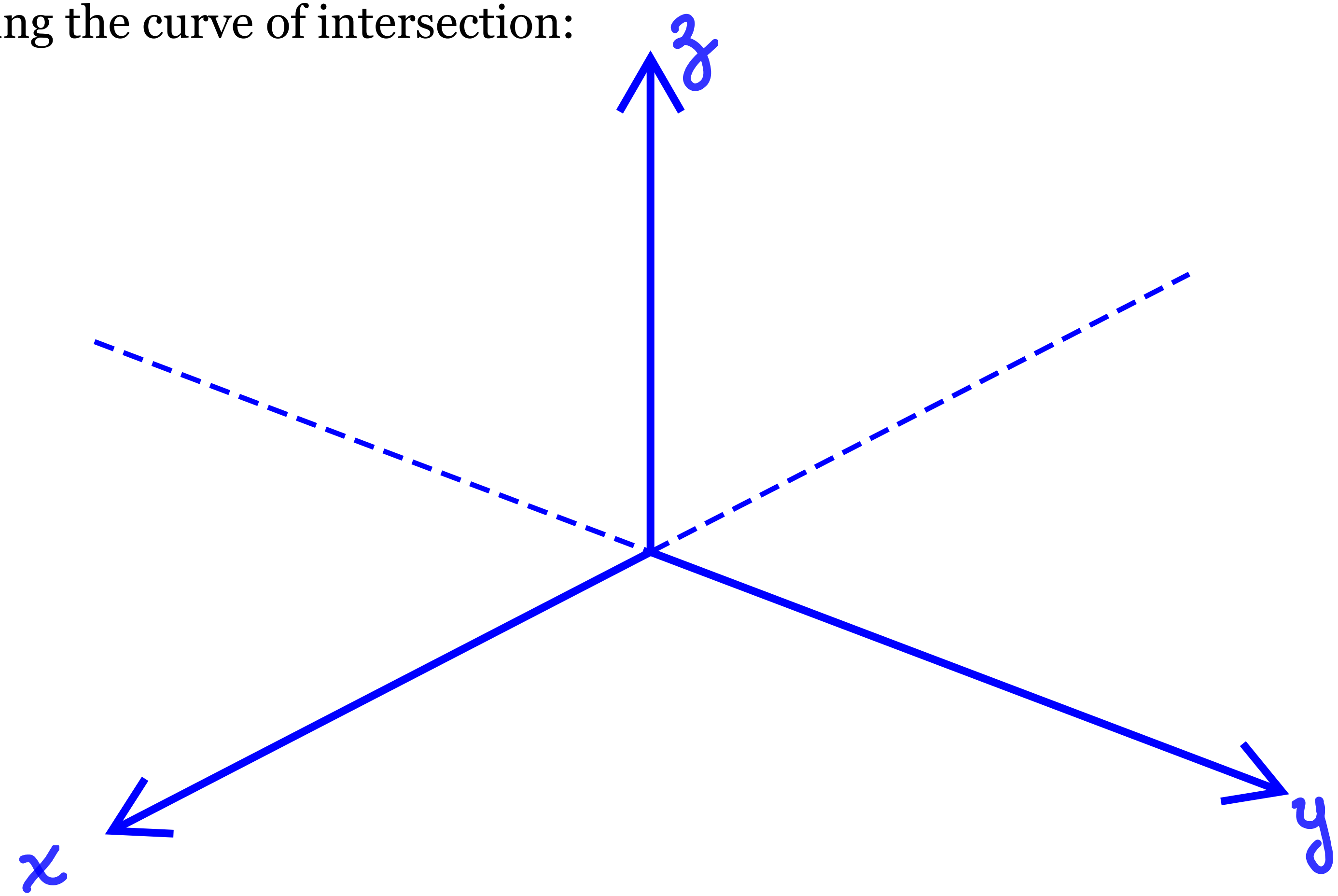
To draw the curve of intersection between surfaces, we need to be able to draw the surfaces and visually determine their intersection.

The surface $z = x^2 + y^2$ is a …

The surface $x + z = 1$ is a …

3D Curves of Intersection

Sketching the curve of intersection:



3D Curves of Intersection

Now that we have sketched the curve, let us determine its **projection on the $\boldsymbol{x - y}$ plane**:

The projection of the curve in the x-y plane can be found by equating the equations of both surfaces:

$$z = x^2 + y^2 \qquad (1)$$

$$x + z = 1 \qquad (2)$$

3D Curves of Intersection

This all looks interesting, but how do we write an expression for this curve in 3D space?

Summary

- We introduced several 3D cylinder and quadric surfaces
- We learned how to recognize them and graph them
- We talked about how curves are formed by the intersection surfaces

We will determine the representation of 3D curves using vector functions during our next lecture.

Thank you